POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2019
Programme: BE | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is 'Turing Test' in Artificial Intelligence (AI)? Criticise the performance of the 'Turing Test' to measure the intelligence of the machine. 2+5
   b) Define AI agents. Explain learning agent with block diagram. 3+5
2. a) There are three pegs, labelled A,B and C. There are 3 disks on peg. The top disk has a diameter of 1, the middle disk has a diameter of 2, and the bottom disk has a diameter of 3. There are no disks on peg C. You must move one disk at time and you cannot place a larger disk on top of smaller disk. The problem is to get all of the disks on peg C. 8
      i. Find a representation for the states of this problem.
      ii. Describe all of the operators that might be applied to a state.
   b) You are given two jugs, a 4- gallon one and a 3 gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug? Solve by production rule system. 7
3. a) Describe the following features of Genetic Algorithms (GA) – Encoding, Fitness Function, Selection, Crossover and Mutation. Discuss how would you encode TSP problem in GA. 7
   b) Describe the behaviour of iterative deepening search in terms of optimality, completeness, time and space complexity. How does it overcome the drawback of BFS. 8
4. a) Prove "You are not doing strawberry picking" using resolution: 8

   If it is sunny and warm day you will enjoy.
   If it is warm and pleasant day you will do strawberry picking
   If it is raining then no strawberry picking.

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If it is raining you will get wet.

It is warm day

It is raining

It is sunny.

b) What is semantic Net? Represent the following fact using partitioned semantic Nets. "Every teacher has taught a student."

5. a) What is Machine Learning and why is machine learning important? Discuss K-Mean Clustering for following dataset with k=2

Staff ID	Age	Nos. of Leaves
1	21	3
2	39	7
3	43	6
4	58	12
5	32	15

b) Assume that the following facts are already entered into the PROLOG database:

```
father(dasharath,ram).    // Dasharath is the father of Ram
father(ram,luv).
father(ram,kush).
mother(kaushalya, ram).   // Kaushalya is the mother of Ram
male(dasharath).          // Dasharath is a male
female(kaushalya).        // Kaushalya is a female
parent(X,Y):-
    father(X,Y);
    mother(X,Y).          // X is a parent of Y
```

6. a) What is Artificial Neural Network? State and discuss backpropagation training algorithm to realize two input XOR gate.

b) What do you mean by NLP? Construct a parse tree for the sentence "The burglar robbed the apartment".

7. Write short notes on: (**Any two**)

a) Reinforcement Learning

b) Deep Learning

c) Expert System

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